Submitted: 06.05.2025 22:00

Applicant	Site of research and department
Maarit Korpi-Lagg	Aalto University, Department of Computer Science Aalto korkeakoulusäätiö Sr., 2228357-4
Call	**Scientific Council**
2025 Special funding for international collaboration in high-performance computing 02.04.2025 - 07.05.2025 16:15	RY

Research topic/Project title

Turbulenttisten ja magnetoituneiden plasmojen mallinnus tekoälyavusteisilla simulaatiomalleilla (MomEnt)
Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)

Funding period	Funding to be applied for	Overall cost estimate
01.01.2026 - 31.12.2028	455 136 €	650 195 €

Keywords

Turbulent Plasma; Magnetised Plasma; Exascale Computing; AI-assisted Simulation; High-performance Computing; Numerical Modelling; Solar Magnetism; Solar Dynamo

Turbulent Plasma; Magnetised Plasma; Exascale Computing; AI-assisted Simulation; High-performance Computing; Numerical Modelling; Solar Magnetism; Solar Dynamo

Field of research

1. Fluid and plasma physics
2. Solar, stellar and interstellar matter physics
3. Computational science
4. Computational data analysis
5. Computer science

Abstract

We develop and use AI-assisted simulation tools to model magnetized plasmas, that occur nearly everywhere in the universe (from galaxies to stars and planets) but are also at the core of many engineering applications (such as fusion reactors). The typical present-day tools use a direct numerical simulation (DNS) approach, attempting to capture all relevant length and time scales of the system in question. It severely suffers from prohibitive computational costs and results in huge output data volumes. It will be very challenging and expensive to go forward with these methods and advance science and industrial applications with the pace required by the society.

To address these challenges we engage artificial intelligence -based methods to the masses of high-fidelity data from the DNS approach, to learn accurate and generalizable descriptions of the small-scale processes. Such descriptions are denoted as sub-grid scale (SGS) models. Our novel simulation tools are built on these SGS models, hence no longer needing to resolve the smallest-scale features and thus reducing to computationally much cheaper models, which require significantly less data storage. These tools are fully scalable to Exascale computing, and enable faster and more sustainable progress in various areas of science and engineering. A proof-of-concept study has been conducted to prove the feasibility of the simulation tool in a simple setting.

This project aims at applying the new simulation framework to study an important real-world application, namely convection under solar-like conditions.
Such a setup acts as a hydromagnetic dynamo, generating and maintaining small- and large-scale magnetic fields. This system has been studied over 70 years without a definite answer to the question how the Sun generates its oscillatory magnetic field and hence its activity cycle. We hope to shed significant new light to the problem with the new simulation tool with unprecedented-resolution numerical simulations, utilising LUMI-G supercomputing infrastructure.

Applicant	Sex	Date of birth	Nationality
Maarit Korpi-Lagg	Female	02.11.1974	Finland

Site of research and department	Present position
Aalto University, Department of Computer Science	Associate professor

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PL 15400 Konemiehentie 2 00076 Aalto Finland	maarit.korpi-lagg@aalto.fi
	Mobile phone 050 4301059
	Telephone, other

Degrees

University of Oulu	Doctor	Astronomy	01.09.1999
University of Oulu	Licentiate	Astronomy	01.09.1998
University of Oulu	Master's degree	Astronomy	01.12.1996

Adjunct professorships

University of Helsinki	Astronomy	01.11.2008

Professorships

Aalto University Associate professor (tenured)	Tietotekniikka	01.01.2020 -
Aalto University Adjunct professor	Astroinformatiikka	01.06.2016 - 31.12.2020
Max Planck Institute for Solar System Research W2 professorship, non-tenured	Aurinkotutkimus	01.06.2016 - 31.08.2021

Public description of project

Nontechnical description in English

We develop and use AI-assisted simulation tools to model magnetized plasmas, pervasive in the universe from galaxies to stars and planets, and core to engineering applications like fusion reactors. Present-day tools employ direct numerical simulation (DNS), aiming to capture all relevant scales but facing prohibitive computational costs and vast data output.

To tackle this, we apply artificial intelligence to DNS data to learn accurate, generalizable sub-grid scale (SGS) models of small-scale processes, reducing computational costs and data storage needs. Our tools are scalable to Exascale computing, enabling faster, more sustainable progress in science and engineering. A proof-of-concept has confirmed their feasibility.

This project aims to apply our framework to study solar convection, that drives Sun's magnetic field generation, unresolved over 70 years. Our simulations, using LUMI-G supercomputing, hope to shed new light on this problem by using unprecedented resolution.

Nontechnical description in Finnish or Swedish

Kehitämme ja käytämme tekoälyavusteisia simulointityökaluja magneettisten plasmojen mallintamiseen, jotka ovat yleisiä maailmankaikkeudessa ja keskeisiä tekniikan sovelluksissa. Aikaisemmat työkalut käyttävät suoraa numeerista simulointia (DNS), pyrkien ottamaan mukaan kaikki asiaankuuluvat mittakaavat, mutta laskentakustannukset ja datan määrä ovat kohtuuttomat.

Innovaatiomme on soveltaa tekoälymenetelmiä DNS-dataan oppiaksemme tarkkoja ja yleistettäviä pienemmän mittakaavan prosessien malleja, mikä vähentää laskentakustannuksia ja datan varastointitarpeita. Työkalumme skaalautuvat eksatason laskentaan, mahdollistaen nopeamman ja kestävämmän kehityksen tieteen ja tekniikan alalla.

Sovellamme uutta simulaatiomenetelmää auringon konvektion tutkimiseen. Tämä mekanismi saa aikaan Auringon magneettikentät, mutta mekanismin yksityiskohdat ovat edelleen mysteeri. Käytämme LUMI-G supertietokonetta, ja etsimme vastauksia simulaatioilla ennennäkemättömällä tarkkuudella.

Salary of principal investigator

Justify in the field below why you include the PI's salary for project management and/or thesis supervision in the total project costs.
Justify in the field below why you apply for funding for the PI's salary for research.
Describe the salary in the field below if the PI does not have an employment relationship for the duration of the funding period.

Collaborators

Project collaborators

Arya Farahi (arya.farahi@austin.utexas.edu)	United States
University of Texas	
Niki Loppi (nloppi@nvidia.com)	Finland
NVIDIA	
Keijo Heljanko (keijo.heljanko@helsinki.fi)	Finland
University of Helsinki	
Hong-Linh Truong (linh.truong@aalto.fi)	Finland
Aalto University	

Mobility

Mobility plan

Name	Duration (months)	From	To
How mobility ties in with the project			
Postdoctoral researcher	2.0	Finland,	United States,
Postdoctoral Researcher		Aalto	University of Texas
Visits to CosmicAI institute; collaboration with Prof. Farahi and other members of the institute.			

Affiliations

National research infrastructures and/or ESFRI roadmap infrastructures

IT Centre for Science	CSC RI
Finnish Grid and Cloud Infrastructure	FGCI
Partnership for Advanced Computing in Europe	PRACE
Partnership for Advanced Computing in Europe	PRACE Suomi

Centre of Excellence

Flagship

Brief data management plan

Brief data management plan (max. 2500 characters). Read more https://www.aka.fi/en/research-funding/apply-for-funding/how-to-apply-for-funding/az-index-of-application-guidelines2/data-management-plan/data-management-plan/

An outcome of this work is releasing our algorithms as a software package via the GitHub repository. Any other code and/or software that is produced as a part of this work will be released through GitHub, including all the ML experiments and evaluations.

New training data will be generated as part of this proposal. The simulation data will be hosted partially on CSC servers and IDA FairData service in Finland. This proposal will generate ML simulation data and associated post-processing data that are of scientific value. These data will be hosted on a public repository and will be released with the codes at the end of the project.

Final peer-reviewed journal manuscripts and supplemental information, such as data tables for graphical information in manuscript figures and for statistically processed averages will be posted in the project digital repository. Authors will ensure their publishing agreement allows the paper to be posted to the arXiv; alternatively, the project website will provide a reference to the journal articles coupled with the supplemental information.

Research ethics

Ethics statement

Ethical permit	Not required
Have you been found guilty of research misconduct during the last three years?	No
Guilty of illegal employment within the meaning referred to in section 7(2) of the Act on Discretionary Government Transfers (688/2001)	No

Progress report

Decision number	Project title	Call	Finance period
359462	Investigating COSmic MAGnetism with GPU-accelerated codes (COSMAGG)	Finansiering för mobilitetssamarbete med Kina, Taiwan eller Tyskland	01.04.2024 - 31.03.2026

Progress report

COSMAGG project was approved by the Research Council of Finland on the 26th of February, 2024. We accepted the funding on the 4th of March, and on the same day Aalto University gave its organizational commitment. In Taiwan, the acceptance was announced only after mid March.

Immediately after the acceptance of the project on both sides, we started bi-weekly telecons between the project partners.
In these telecons we have advanced the collaborative work remotely, but also the workplan and timetable of the collaborative visits.
The first visit took place in June 2024 by doctoral student Johannes Pekkilä, to visit Taiwanese partners for a month.
Work conducted and completed during the visit was the implementation of spherical coordinates and non-equidistant grids to the Astaroth library.

During February, 3 colleagues from ASIAA visited Aalto for three weeks. We implemented additional physics relevant for the studies of the interstellar matter and other compressible fluids to the Astaroth library. Currently our doctoral student Touko Puro is visiting ASIAA to finalise these implementations.

These improvements can also indirectly benefit the MomEnt project project in terms of the enhanced stability of the solver when shock fronts develop near the surfaces of stellar convection zones. Otherwise there in no significant overlap of COSMAGG and MomEnt projects.

Most relevant publications and other key outputs

Publications

Year of publication	Authors of publication	Name of publication	Name of journal/series	Permanent web address
2025	Touko Puro, Matthias Rheinhardt, Maarit Korpi-Lagg, Hong Linh Truong, Minh Tri Nguyen, Anh Dung Nguyen	In-situ and Opportunistic Data Analysis for Exascale Astrophysics Simulations	Computer Physics Communications	https://github.com/ToxPuro/In-situ-and-Opportunistic-Data-Analysis-for-Exascale-Astrophysics-Simulations
2024	Yokelson, Dewi; Lappi, Oskar Johannes; Ramesh, Srinivasan; Väisälä, Miikka; Huck, Kevin; Puro, Touko; Norris, Boyana; Korpi-Lagg, Maarit; Heljanko, Keijo; Malony, Allen D.	SOMA : Observability, monitoring, and in situ analytics for exascale applications	Concurrency and computation: practice and experience	http://dx.doi.org/10.1002/cpe.8141
2024	Nguyen, Minh Tri; Nguyen, Anh Dung; Rantaharju, Jarno; Puro, Touko; Rheinhardt, Matthias; Korpi-Lagg, Maarit; Truong, Hong Linh	Supporting Opportunistic Data Operations for Data-Intensive Computational Applications	IEEE International Conference on Big Data	
2023	Warnecke, Jörn; Korpi-Lagg, Maarit J.; Gent, Frederick A.; Rheinhardt, Matthias	Numerical evidence for a small-scale dynamo approaching solar magnetic Prandtl numbers	Nature Astronomy	
2023	Rempel, Matthias; Bhatia, Tanayveer; Bellot Rubio, Luis; Korpi-Lagg, Maarit J.	Small-Scale Dynamos : From Idealized Models to Solar and Stellar Applications	SPACE SCIENCE REVIEWS	
2022	Pekkilä, Johannes; Väisälä, Miikka S.; Käpylä, Maarit J.; Rheinhardt, Matthias; Lappi, Oskar	Scalable communication for high-order stencil computations using CUDA-aware MPI	PARALLEL COMPUTING	
2021	Väisälä, Miikka; Pekkilä, Johannes; Käpylä, Maarit; Rheinhardt, Matthias; Shang, Hsien; Krasnopolsky, Ruben	Interaction of large- and small-scale dynamos in isotropic turbulent flows from GPU-accelerated simulations	Astrophysical Journal	
2021	Brandenburg, Axel; Johansen, Anders; Bourdin, Philippe; Dobler, Wolfgang; Lyra, Wladimir; Rheinhardt, Matthias; Bingert, Sven; Haugen, Nils; Mee, Antony; Gent, Frederick; Babkovskaia, Natalia; Yang, Chao-Chin; Heinemann, Tobias; Dintrans, Boris; Mitra, Dhrubaditya; Candelaresi, Simon; Warnecke, Jörn; Käpylä, Petri; Schreiber, Andreas; Chatterjee, Piyali	The Pencil Code, a modular MPI code for partial differential equations and particles: multipurpose and multiuser-maintained	JOURNAL OF OPEN SOURCE SOFTWARE	http://adsabs.harvard.edu/abs/2021JOSS....6.2807P
2018	Warnecke, Jörn; Rheinhardt, Matthias; Tuomisto, Simo; Käpylä, Petri J.; Käpylä, Maarit J.; Brandenburg, Axel	Turbulent transport coefficients in spherical wedge dynamo simulations of solar-like stars	Astronomy and Astrophysics	http://adsabs.harvard.edu/abs/2016arXiv160103730W
2017	Pekkila, Johannes; Vaisala, Miikka S.; Käpylä, Maarit J.; Käpylä, Petri J.; Anjum, Omer	Methods for compressible fluid simulation on GPUs using high-order finite differences	Computer physics communications	

Justification for selected publications

These publications manifest
- excellent science conducted by the research group in the field of astrophysical fluid dynamics
- the efficient tools that the research group has produced, that have enabled us to reach world-leading level in astrophysical fluid dynamics
- the intensive and fruitful collaboration that has been established with the listed project partners

Other key outputs

Output	Name of output
developed practises, tools and software	Astaroth GPU library
developed practises, tools and software	Pencil Code
developed practises, tools and software	ODOP

Motivation for selected outputs

Astaroth is a multi-GPU library for stencil computations for multiphysics simulations. It is the main computational engine used in the HPC pipeline: https://libastaroth.bitbucket.io/

Pencil Code is a high-order finite-difference code for compressible hydrodynamic flows with magnetic fields and particles parallelized on CPUs utilizing MPI and multi-threading through openMP. Astaroth is interfaced to Pencil-Code to achieve efficient performance on CPUs and GPUs: https://github.com/pencil-code

ODOP is a framework supporting the developer/scientist to explore free resources allocated for their long-running applications to carry out data operations in an opportunistic manner: https://github.com/rdsea/odop/

PROJECT FUNDING

Total costs

Salaries	2026	2027	2028	Total
Postgraduate student	35 200	35 904	36 619	107 723
Postdoctoral Researcher	45 650	46 563	47 498	139 711
Indirect employee costs 46.0 %	37 191	37 935	38 694	113 820
Overheads share 75.0 %	88 531	90 302	92 108	270 941
SALARIES TOTAL	206 572	210 704	214 919	632 195

Other costs (incl. VAT)	2026	2027	2028	Total
Travel expenses	6 000	6 000	6 000	18 000
OTHER EXPENCES TOTAL	6 000	6 000	6 000	18 000

Total project costs

Total	2026	2027	2028	Total
	212 572	216 704	220 919	650 195

Funding received from other sources

Funding sources	2026	2027	2028	Total
Own organisation	63 772	65 011	66 276	195 059
Total	63 772	65 011	66 276	195 059
Other funding (%)	30.00	30.00	30.00	30.00

Funding to be applied for from the Research Council

Total	2026	2027	2028	Total
	148 800	151 693	154 643	455 136
Research Council funding contribution (%)	70.00	70.00	70.00	70.00

Justification

PhD student: a computer science PhD student. Postdoc: application domain science postdoctoral researcher

Appendices

Heading	File	Date
Complete list of publications	Korpi_Lagg_publ_wc.pdf	06.05.2025
Letter of collaboration	support_letters.pdf	06.05.2025
CV	Korpi_Lagg_CV_3page.pdf	29.04.2025

Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools

Maarit J. Korpi-Lagg (MomEnt)

1 Background and significance

1.1 Significance of research project in relation to current knowledge, research-based starting points

Precise and accurate modeling of magnetized plasmas stands as a formidable challenge that has long vexed a broad spectrum of the scientific community, ranging from reactor engineers and space scientists to astrophysicists [3, 24, 36, 6, 2]. This challenge stems from the inability to run direct numerical simulations (DNS), which can faithfully represent both small-scale and large-scale physics due to the spatial and temporal scales that cover many orders of magnitude as well as the extreme conditions prevalent in these environments. Even with the current supercomputing resources, we are able to run only DNS-type models, where the values of diffusivities are elevated by orders of magnitude from their real values. An example of a high-resolution DNS-type simulation produced by our group is shown in Figure 1.

Mean-field models have emerged as a more accessible alternative by simplifying the representation of smaller-scale phenomena through turbulent transport coefficients that couple small-scale physics with large-scale physics [20]. However, their efficacy is compromised by a fundamental gap in our understanding of turbulent transport mechanisms, leading to the reliance on overly simplistic or selectively chosen turbulent effects [7, 46, 40].

To address these challenges, subgrid-scale (SGS) models[1] have emerged as a popular and effective alternative. Implicit Large Eddy Simulations (ILES) employ a class of numerically motivated SGS models, designed for simulating plasmas in a manner maximizing the flow Reynolds numbers, which controls the intensity of turbulence [for a recent review, see 22]. Despite its success in many applications [41, 30], ILES may not be fully suitable for, e.g., solar-like complicated magnetised plasmas [39], where significant differences in behavior have been observed when compared to DNS-type simulations [15, 17, 14, 43]. These discrepancies highlight the need for more sophisticated SGS models. They have been sought with ansatzes stemming from mean-field angular momentum and electrodynamics theories. One

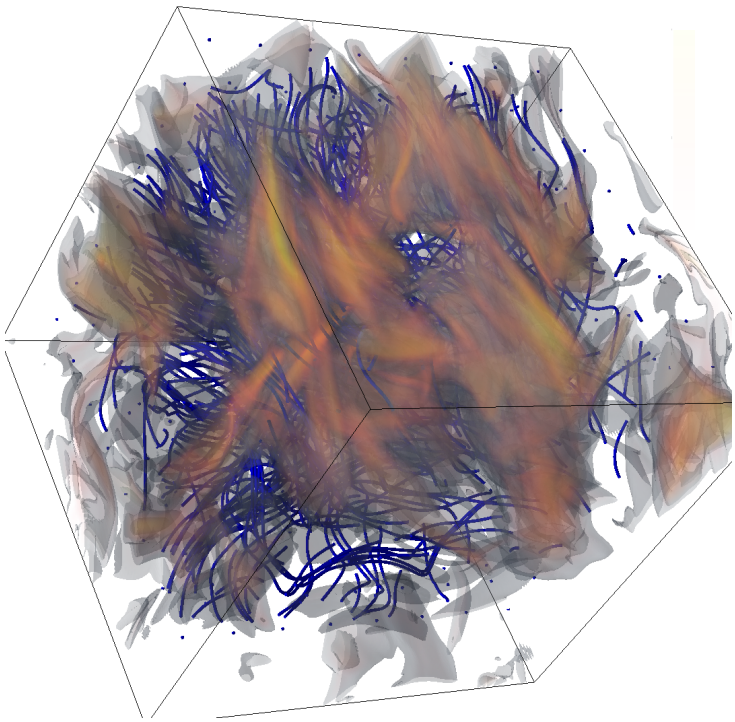

Figure 1: Dynamo-active isothermal forced flow simulated with hybrid MHD simulation tool PC-A. These simulations of the generation and saturation of magnetic field fluctuations, through a so-called small-scale dynamo instability, represent world's largest simulations of their kind [44].

[1]Subgrid-scale models are surrogate models that bridge the gap between large-scale flow behavior and small-scale processes in simulations by approximating the effects of smaller scales, which remain unresolved due to computational constraints. These models aim to provide a sufficiently accurate representation of small-scale physical phenomena, such as turbulent transport, by incorporating their cumulative impact on the larger scales.

approach fits the coefficients to the simulated data using techniques such as singular value decomposition and enhancements [5, 29, 38, 1, 37], another using test field methods [34, 35, 4, 11, 45, 46, 19]. The former suffers from the non-uniqueness of the fitting procedure and hence produces unreliable results [16], and the latter becomes unpracticable for large-scale data due to the huge computational resources required.

The most sophisticated framework proposed so far is the Explicit LES (ELES) approach, which employs physically motivated SGS models to capture better the complexity of flows affected by rotation, density stratification, and magnetic fields. Some ELES-SGS frameworks have been proposed [13, 12], but none of these methods have been rigorously shown to capture the turbulent transport processes that lead to large-scale structure formation in realistic settings, such as the generation of global-scale magnetic structures and oscillatory dynamo modes in the Sun.

Artificial intelligence (AI) and machine learning (ML) models have recently been increasingly employed to address the computational intensity and complexity of traditional simulation methods [10, 18, 42]. These techniques offer significant speedups in computation time and the ability to capture complex physical phenomena through data-driven models [9, 32, 33]. By training on data from DNS-type and even empirical data, these models have the potential to infer underlying physical processes at small scales and reveal new insights into the behavior of plasmas. Despite recent advances, how to model and capture small-scale physics and create effective and scalable AI-assisted ELES models for turbulent and magnetized plasmas remains an open problem.

We propose a novel path forward, solving all the problems described above:

- **GPU-DNS concept**: We will take advantage of GPU-accelerated DNS-type simulations (see Figure 1), enabling the production of unprecedented resolution simulations using the fastest available supercomputer resources, such as the Large Unified Modern Infrastructure (LUMI) and its GPU partition in Finland. These methods have been developed and optimized in-house in our research group, and represent cuttings edge performance and scale-up of high-order turbulence simulations in the world.
- **AI-IN-LOOP concept**: We will undertake the data analysis of the simulations while they are running (known as a 'simulation-in-the-loop' approach), streaming the data between the DNS-type simulation and AI tool, without any necessity to dump the huge system states to the disk, completely avoiding the data analysis bottleneck. This way, we will train deep neural networks (DL) to learn accurate and generalisable descriptions (hereafter denoted subgrid-scale, SGS, models) of the small-scale processes on-the-fly.
- **LES-DNS concept**: We will build next-generation simulation tools on the derived SGS models. SGS models aim to bridge the gap between large-scale behaviour and small-scale processes by approximating the effects of the smaller scales. This project aims at applying the new simulation framework to study an important real-world application, namely convection under solar-like conditions. Such a setup acts as a hydromagnetic dynamo, generating and maintaining small- and large-scale magnetic fields. This system has been studied over 70 years without a definite answer to the question how the Sun generates its oscillatory magnetic field and hence its activity cycle driving space weather phenomena.

Our next-generation simulation tools aim to provide a sufficiently accurate representation of small-scale physical phenomena by incorporating their cumulative impact on larger scales. As a result, these models will no longer need to resolve the smallest-scale features, reducing to computationally much cheaper models and therefore require significantly less data storage. These tools will enable faster and more sustainable progress in various areas of science and engineering. Our research group has the required expertise to carry out this project in all aspects. The HPC technologies required to undertake the project have been developed in house (with support from two ERC grants and Academy of Finland Center of Excellence, but all for astrophysics research) Our

group members are the world-leaders of modelling the solar magnetic activity We have developed the SGS and LES approaches of magnetized plasmas over decades with well-documented results. With the support from our collaborators, especially important in developing the AI techniques, we will, with no doubt, be able to successfully carry out this ambitious project and all of its subtasks. In addition to these collaborators, we have an extensive international network interested in utilizing our HPC tools, including research groups from Austria, Germany, India, Sweden, Switzerland, Taiwan, and several groups from the United Kingdom (University of Exeter, Glasgow, and Newcastle Upon Tyne), to be added with industrial collaborators.

1.2 Research questions and/or hypotheses

Our main hypothesis is that by creating an AI framework for extracting actionable insights from high-fidelity data[2] we can formulate a physically plausible SGS model. Based on it, we further hypothesize that we can develop an equally accurate as DNS-type models ELES simulation tools. As they are based on filtered equations, they will form a novel, computationally cheaper, class of simulation tools. This framework aims to capture all the relevant small-scale physics associated with a turbulent hydromagnetic environment including the physics-driven coefficients describing the turbulent transport in such systems. In particular, the inherent inhomogeneities and anisotropies occurring at these scales have to be accounted for, as well as further anomalies that have been notoriously difficult to model. Our hypotheses can be mathematically formulated as follows. Specifically, we will replace the equations nominally used to describe the bulk behaviour of plasma (know as the magnetohydrodynamic (MHD) differential equations) – solved in DNS – by a filtered formulation of the key plasma properties in this physics setup.

$$\frac{\partial \overline{\boldsymbol{B}}}{\partial t} = \nabla \times \left(\overline{\boldsymbol{U}} \times \overline{\boldsymbol{B}} + \overline{\boldsymbol{\mathcal{E}}}\right) + \eta \nabla^2 \overline{\boldsymbol{B}}, \tag{1}$$

$$\overline{\rho}\frac{D\overline{\boldsymbol{U}}}{Dt} = -\boldsymbol{\nabla}\overline{p} + \boldsymbol{\nabla}\cdot(2\nu\overline{\rho}\overline{\mathsf{S}} - \overline{\boldsymbol{\mathcal{R}}} + \overline{\boldsymbol{\mathcal{M}}}), \tag{2}$$

$$\overline{\rho}\overline{T}\frac{D\overline{s}}{Dt} = -\boldsymbol{\nabla}\cdot(\overline{\boldsymbol{F}}_{\rm rad} + \overline{\boldsymbol{\mathcal{F}}}) + \overline{Q}_{\rm diss}, \tag{3}$$

$$\frac{D\ln\overline{\rho}}{Dt} = -\boldsymbol{\nabla}\cdot(\overline{\boldsymbol{U}} + \overline{\boldsymbol{\mathcal{D}}}). \tag{4}$$

The equations describe the time evolution of the magnetic field (1), momentum (2 Navier-Stokes equation for viscous fluids), entropy (3), and continuity (conservation of mass, 4). The overbar indicates filtering. p is the pressure, ν and η are the microscopic diffusivities, $\overline{\mathsf{S}}$ is the traceless rate of strain tensor, $\overline{\boldsymbol{F}}_{\rm rad}$ is the radiative flux, and $\overline{Q}_{\rm diss}$ represents the dissipative heat sources, while $\overline{\boldsymbol{\mathcal{E}}}$, $\overline{\boldsymbol{\mathcal{R}}}$, $\overline{\boldsymbol{\mathcal{M}}}$,$\overline{\boldsymbol{\mathcal{F}}}$ and $\overline{\boldsymbol{\mathcal{D}}}$ are the electromotive force, Reynolds stresses, Maxwell stresses, turbulent heat flux, and turbulent mass flux, respectively, calculated over the filter scale. The latter five quantities, defined by the correlations of small-scale fields, are the SGS components to be modeled with ML.

Although physics-based models and physical symmetry constraints of the environment (e.g., gravity and rotation) are proposed for many of these components, their effectiveness is limited. Here, the purpose of the AI technique used is to learn new models that remain unchanged in these symmetries. In this part of the project, we aim to expand our initial DL results (see Figure 2) to more complicated scenarios e.g., including more complex physics **This moves beyond more traditional approaches**, with the main goal to develop **evolution equations** for five plasma quantities over the filter scale. The AI methods will then address their non-local and non-linear interactions with the resolved, now filtered, fields [31]. Once fully implemented, these filtered plasma parameters will be

[2]here, high-fidelity refers to data that is created by high-resolution, expensive DNS simulations.

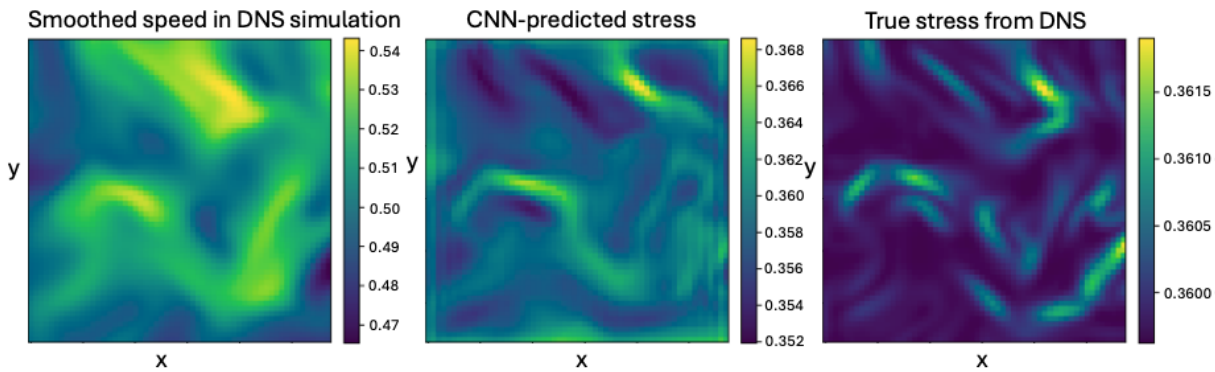

Figure 2: Proof-of-concept Taylor-Green vortex DNS simulation results (smoother mean DNS flow field shown on the panel on the left, one of the generated Reynolds stress components shown in the panel on the right) with the offline-trained AI framework and the reconstructed stresses with a U-Net type Convolutional Neural Network with symmetric upscaling and downscaling blocks and skip connections [21]. Image credit: Niki Loppi, NVIDIA.

compared and evaluated against their DNS-type counterparts. The performance of our novel models will be measured with respect to the computational time and energy consumption compared to current methods. The accuracy of our filtered parameters will also be calculated using conventional approaches (e.g., optimising the root-mean-squared error of parameters), to allow for easy comparison with current models and potential competitors.

1.3 Special objective of call

The MomEnt project addresses the following relevant aspects of the call:

- It utilises the LUMI-G supercomputer for top-notch research in fluid dynamics, relevant for many other scientific fields, such as astro-, space-, and fusion plasma physics. PI's group has actively used the LUMI supercomputer from the very beginning of its history: First we acted as pilot users of LUMI before its inauguration, and now we are the biggest individual group utilizing it from Aalto University. We have received several standard project allocations, and extreme-scale access allocations both through the Finnish and European quota.
- It builds on active collaboration with the NSF-Simons AI Institute for Cosmic Origins (Cosmic AI, https://www.cosmicai.org/), especially with the Artificial Intelligence Team Lead Prof. Arya Farahi from University of Austing, Texas, US. Collaboration with Prof. Farahi concentrates especially on the improvement of the proof-of-concept AI tools to take into account the explainability aspect of them. This is vitally important goal to be able to build physically plausible SGS models with the help of the AI tools.
- It integrates recently developed HPC tools (during ERC grants UniSDyn and SYCOS, and EuroHPC Inno4Scale project NEOSC) into a diverse HPC ecosystem, where all the building blocks together enable high-profile and societally important computational research in pre-exascale supercomputers and beyond. Furthermore, our tools automatically optimize for utilizing the heterogeneous resources as efficiently as possible, thus significantly contributing to sustainable and green computing. Furthermore, the tools and methods developed in this project are perfectly suitable to LUMI AI factory.

As is presented in Figure 3, our HPC workflow scales up in LUMI-G nearly perfectly both in weak and strong scaling scenarios.

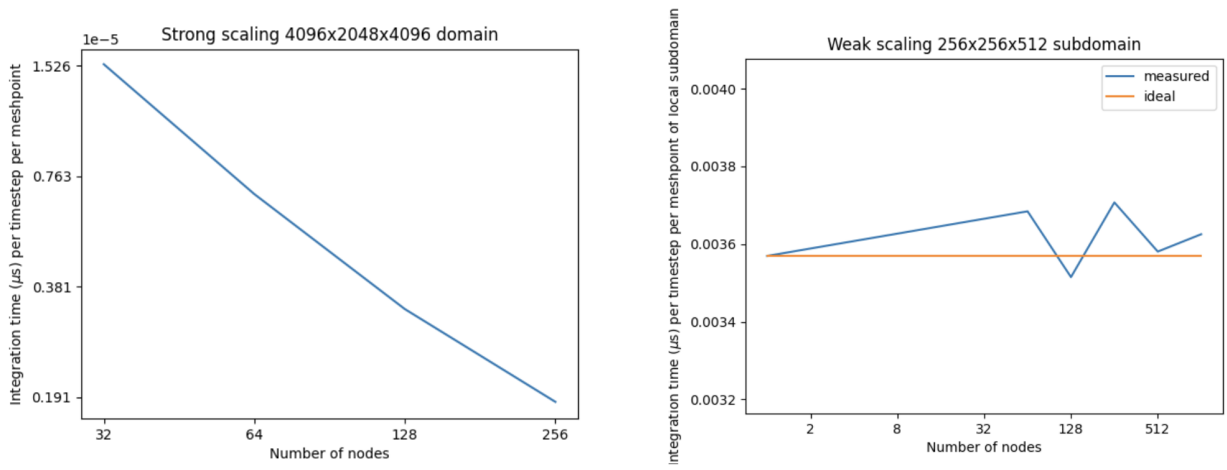

Figure 3: Scaling results of the full HPC workflow (PC-A and AI tools implemented through ODOP)in LUMI-G. The used setup us spherical turbulent convection with a sub-domain size 256x256x512 in the weak scaling case representing the planned simulation runs in the GPU-DNS concept phase [28].

2 Impact

2.1 Outcomes and impacts of research within academia

Our project aims to transform plasma simulation tools to the next level by creating novel AI methods for modeling magnetized plasmas,significantly improving computational speed and reducing computational cost. The utilization of the developed AI methods widens plasma simulations' applicability in research, spanning from solar phenomena to astrophysical and technological domains. Incorporating explainable AI (XAI), we will unlock and clarify complex plasma interactions within these systems, enhancing scientific understanding and innovation. This blend of transformative simulation and AI models, uncertainty quantification methods, and deep insights gained through explainable AI promises a paradigm shift in plasma physics research.

2.2 Effects, impact and interaction beyond academia

This project democratizes complex magnetized plasma simulations, currently limited by high computational costs and storage requirements. This project will engage a broader research community, facilitating the integration of plasma simulations with empirical research and the development of practical applications, such as space weather prediction, which require real-time data and high accuracy. Collaboration between leading international groups in AI methods and plasma simulations will foster a transdisciplinary community skilled in both physics and AI/ML. This workforce is vital for academia and industry alike. Important aspect is to democratize access to AI and plasma physics for diverse audiences, fostering inclusivity and equipping participants with cutting-edge skills essential for future scientific and engineering challenges.

3 Implementation

3.1 Work plan and schedule

Our project developes three novel concepts; the work plan can be divided into phases based on them. A detailed workplan is presented below and summarized in Figure 4.

Months 1–6: Integration of various components, their performance and scalability [GPU-DNS] – PD: development, testing and verification of the solar convection dynamo model in spherical coordinates using PC-A solver and testing and validating the DL concept in simple known scenarios PS: integration, performance, and scalability of PC-A and DL simulation-in-the-loop concept

[AI-IN-LOOP] PI: leading the integration and securing the computing resources at CSC and in LUMI-G.

Months 6–18: Extending the DL framework to magnetised plasmas – PS: Implementation of the novel LES-DNS simulation framework to PC-A (see Section 3.2 for details of the research methods) begins [LES-DNS]. PD: extension and required development and optimisations of the DL framework to the desired solar convective dynamo takes place. PD makes the first visit to University of Texas [AI-IN-LOOP]; PI: leading the novel LES solver implementation, contributing to the AI methods development, and securing further computational resources.

Months 18–24: Full next-generation plasma simulation chain [LES-DNS] is in place, tested, verified, and optimised – PS: verification and optimisation of the [AI-IN-LOOP] production pipelines. Postdoc: testing, verification, and optimisation of the convective dynamo PC-A part of the [LES-DNS] chain. PD: optimisation and further development of the AI part if necessary. PI: verification and optimization of the [LES-DNS] concept, and securing computational resources.

Months 24–36: Production runs with the full novel LES-DNS chain with unprecedented resolution of the solar convection dynamo setup. Analysis of the final results, assessment of the accuracy and validity of the developed [AI-IN-LOOP] concept. Main dissemination of the results. PD: main responsibility for running and analysing the results. PD makes the second visit to University of Texas. PS: dissemination of the methods and securing the production runs operability. PI: participating in the design of the production runs, data analysis, and dissemination of the results.

3.2 Research data and material, methods, and research environment

We will use the following **research methods**:

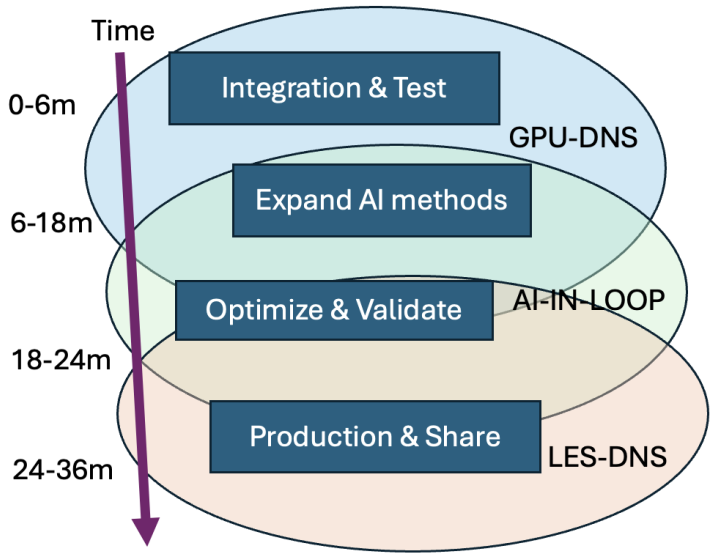

Figure 4: Timetable

- The public domain stencil library Astaroth will be the main computational engine employed during the project [26, 25].
- Astaroth is interfaced to Pencil Code, which both orchestrates the GPU computations and performs some recurrent diagnostics on the fly, utilising the CPU cores that are not involved in stencil computations through multithreading with OpenMP (PC-A; [27]).
- The ODOP framework [23] is utilised to perform further data analysis and movements, maximising the resource usage on the reserved nodes, optimising for the quality of the data products, and minimising the data storage needs on the HPC system.
- The AI development platform used is NVIDIA's PhysicsNeMo, an open-source framework for physics-based AI models. This framework uses various techniques needed to achieve our objective including, but not limited to, physics-informed neural networks and Fourier neural operators to create an architecture for analyzing MHD and differential equations as well as plasma data
- NVIDIA Warp framework will be used to build accelerated and differentiable simulation code.

- Additionally, NVIDIA's TorchFort library will be used to integrate our DL model with the DNS tool (PC-A), enabling real-time training and inference. We will apply both physical and numerical constraints to ensure the generalizability of the DL model.

The main **research environment** is the Department of Computer Science (hereafter Aalto/CS) at the Aalto University School of Science. With over 40 professors and more than 400 employees, the department is the largest Computer Science unit in Finland. It is ranked among the top 10 Computer Science departments in Europe and among the top 100 globally. ICT and digitalization are among the seven key research areas at Aalto University. In addition to standard resources provided by the department, there is support for midrange HPC and data analytics through the Science-IT project of the Aalto University School of Science. Science-IT and its flagship HPC unit Triton, are a part of the national Finnish Computing Competence Infrastructure (FCCI). Currently, Triton provides 9600 computing cores and over 30 servers for high-end GPU computing. For indepth research methods and software development, Science-IT provides support from research software engineers. Researchers also have free access to national CSC resources, including the supercomputers Puhti and Mahti, as well as the cloud resources Rahti and CPouta. The teams also have access to Europe's most powerful GPU-based supercomputer LUMI through competitive computing time allocations via the Euro-HPC consortium. The consortium has an outstanding track record of obtaining competitive computational resources through the Partnership for Advanced Computing in Europe (PRACE), EuroHPC, and CSC.

We produce mainly **simulated data** in various formats (HDF5 and binary/ASCII files), Fortran or C binaries, and AI models in suitable formats. All formats for data are highly standardised and described in the Pencil Code and Astaroth user manuals. A minimum amount of system states (in binary or HDF5 format) will be kept for allowing re-starts of the most important models. The main emphasis is on performing various types of data analyses and reductions on the fly (for example, various kinds of averages or power spectra) to provide data in a more compact and, hence, more manageable form, especially when high cadences are needed.

3.3 Risk assessment and alternative implementation strategies

This is a pioneering effort to use AI and DL for describing highly non-linear and non-local processes, and is therefore of high risk. If computationally affordable and sustainable simulation tools can indeed be formulated and developed this way as we hypothesize, this will mark a very significant milestone in computational science and engineering in general.

To achieve our objective using our AI implementation, there are many risks that need to be considered and mitigated. The main risk that can hinder this project is the potential for delays in securing computing resources. We can mitigate such delays with our already secured initial allocations of GPU time. Throughout the three-year project we will proactively apply for additional resources to ensure uninterrupted progress.

Minor risks that could potentially be an obstacle include the complexity of developing this state-of-the-art stable and generalisable SGS model for magnetised plasmas. To address this, we will adopt an iterative approach, allowing flexibility in model adaptation without compromising the project's overall objectives. Additionally, the prescribed LES approach is expected to yield significant speedups, making simulations more computationally feasible.

PI's group has extensive, world-leading expertise in high-performance computing and plasma modelling, ensuring that we can effectively handle large-scale data and computational demands. The GPU-optimised solver (Astaroth) is already in production, reducing implementation risks, and currently providing an order of magnitude performance improvement when compared to previous CPU methods. The methodologies we develop will not only support this project but also provide valuable tools for the broader plasma science community, ensuring long-term impact beyond the

immediate research goals.

3.4 Project personnel and their project-relevant key merits

The PI, **Maarit Korpi-Lagg** (over 4,800 citations and h-index of 39 according to Google Scholar), is a tenured associate professor at Aalto CS, where she leads the High-Performance Computing Lab. She was the grant holder of the ERC Consolidator grant "UniSDyn" and the ERC Proof of-concept grant “SYCOS”, which projects studied dynamo action in solar and stellar convection zones. She acted as a team leader of the Academy of Finland Centre of Excellence “ReSoLVE” (2014-2019), which studied long-term solar magnetic activity and its effects on Earth. She has acted as a PI of several competitive HPC projects (LUMI-C and LUMI-G Finnish project allocations, PRACE, DEISA/DECI, CSC Grand Challenge), and is a core developer of the Pencil Code Multiphysics Solver [8] for high-order stencil-based applications, and the initiator and project leader of the GPU-based MHD solver Astaroth.

The main research objective of our HPC Lab for the past five years has been to develop an efficient GPU-accelerated DNS tool for modelling magnetized plasmas. This project is nearing completion [26, 25, 28], with the DNS tool is in full production use in the LUMI-G partition. It has also been ported and passed technical inspection in various other supercomputers across the globe. We have been successful in obtaining large amounts of computing time at various sites. The catch here is that, while we can now go to unprecedented resolutions from the compute point-of-view, the amount of data produced by the simulations is growing too large (soon approaching Petabyte scale) to be stored in the HPC systems. In practise, the exploding amounts of data prohibit us to produce simulations at the pace we otherwise could. This kind of approach to computing will face a dead-end in the near future. Our HPC Lab has developed tools to overcome the data analysis bottleneck challenges, e.g. utilizing streaming of data from the memory of the main application of the data analysis tools on the fly, and these will be utilized here.

All these computational (and therefore sustainability) challenges we are currently facing have led us to consider the novel concept of engaging machine learning to make use of the masses of data that we currently produce, and to develop this way the next-generation modelling tools that utilize the learnt SGS models to infer the smallest-scale effects. We successfully obtained seed funding from the FARIA Global US Networking Fund of the Ministry of Education and Culture. During the project, we have actively searched for collaborations in AI, and this is now bearing fruit. For example, we have now established collaboration with the CosmicAI NSF-Simons Foundation Centre through Prof. Arya Farahi (University of Texas at Austin, coordinator of CosmicAI). We also applied for a collaborative project from the NVIDIA AI Technology centre and were awarded three person months of research engineer support to develop and apply the necessary AI methodology. The initial deep convolutional neural network has been developed, trained, and tested with data dumped to disk from a low-resolution fluid simulation with more simple physical processes. The LES-framework for the new simulation toolbox has been designed, but for its implementation, testing and verification we need further personnel resources.

The project personnel will consist of a postdoctoral researcher (PD) and a postgraduate student (PS). The **PD, to be recruited**, must have a doctoral degree in astrophysics and a strong background in computational science. The PD’s tasks include running the simulations and developing, supervised by the PI, the SGS models and the new LES solver. The PD will collaborate closely with Prof. Farahi to transform the AI tool data into knowledge that can be formulated into SGS models. **The PS in computer science has been recently recruited (MSc Touko Puro)**; the thesis topic is "Heterogeneous Exascale Scientific Computing Pipeline with Domain-Specific Language Compiler Techniques". Touko’s tasks shall include all the development and optimization tasks related to the simulation, AI, and data analysis tools, and the construction of the new LES solver. Touko is already well-integrated into our HPC lab and is intensively collaborating with Prof. Truong’s group, Dr. Loppi,

and Prof. Heljanko's team at UH.

3.5 Collaborators

Prof. Arya Farahi, University of Texas, Austin, Tx, US: Prof. Farahi is an Assistant Professor in the Statistics and Data Science department, and his research is focused on advancing research at the intersection of artificial intelligence, machine learning, and scientific applications. His expertise lies in developing explainability and uncertainty quantification methods for AI models, particularly those designed for complex scientific domains such as astronomy and solar physics. His support for the project is vital in the development of explainable AI methods and applying them to complex astrophysical flows.

Prof. Keijo Heljanko, Department of Computer Science, University of Helsinki: Prof. Heljanko is a Professor of Parallel and Distributed Data Science. He is researching Big Data processing using parallel algorithms that employ both CPUs and GPUs. He is the director of the Data Science Master's Program at the University of Helsinki and FCAI representative in the joint technology center of NVIDIA and FCAI (NVAITC) accelerating research, education and adoption of artificial intelligence in Finland. During the most recent years, he has worked intensively on transforming different types of Big Data processing algorithms to parallelize efficiently on GPUs. Prof. Heljanko will crucially support this project in the extremely challenging data analysis tasks and workflows.

Dr. Niki Loppi is an AI/HPC solutions architect at **NVIDIA Helsinki**, helping academic researchers to leverage NVIDIA's technology stack through the NVIDIA AI Technology Center program. Prior to joining NVIDIA, he worked as a researcher in the Department of Aeronautics at Imperial College London, where he also obtained his PhD in Computational Fluid Dynamics. Specifically, his research focused on the development of high-order accurate numerical methods for solving large-scale incompressible fluid flow problems using massively parallel modern GPU architectures. Currently he works on physics-informed neural networks and scientific AI in general. Dr. Loppi has collaborated with the PIs group intensively during the past year, and he has developed the proof-of-concept U-net type CNN model, which we have successfully applied in collaboration to the case of Taylor-Green vortex (see Figure 2). Dr. Loppi will collaborate with the PI in identifying of the most effective neural network architecture for MHD sub-grid scale modeling.

Prof. Hong-Linh Truong, CS, Aalto University: Prof. Truong leads the group of Systems and Service Engineering Analytics (AaltoSEA) engaged in research and development of concepts and techniques for big data systems, resource management, performance monitoring, and big data analytics for complex software systems. Prof. Truong has participated in several EU projects involving HPC, Grid computing, and scientific workflows. His role in this project is to contribute to the integration of cloud services to the HPC ecosystem, specifically to the implementation of profiling and service integration, and ecosystem-aware experiment management and explainability. The PI and prof. Truong have collaborated intensively for several years. This collaboration has resulted in the development of the ODOP tool, which will be a key component to be used in the simulation work and developed further during this project.

3.6 Responsible science

There are **no ethical concerns** as the application deals with no sensitive data. Nevertheless, we will follow stringent integrity principles in accordance with the guidelines of the Finnish Advisory Board on Research Integrity, and the ALLEA European Code of Conduct for Research Integrity.

We are committed to promote **diversity, equality and non-discrimination** in all our activities. Statutory obligations set forth in the Act on Equality between Women and Men 609/1986, 1329/2014 and the Non-discrimination Act 1325/2014 also apply to our plan. We are firmly committed to promote equal opportunities for individuals to learn, acquire knowledge, participate

and make a difference. We will make our values visible in all our practices.

The products of the research will be made available immediately to the community during the project. **Fully open access publications, data, and source code policy will be adopted.** Services like Zenodo and FAIRdata IDA will be utilised for opening up data products ranging from the data to reproduce all the figures, to intermediate and selected raw data products for reproducibility and reusability of the results. All the simulation and data analysis tools are developed as open source, available and continuously maintained under GitHub and Bitbucket repositories.

Our simulation and data analysis tools are being developed to work generally on GPUs and hence are energy-efficient in comparison to standard CPU computing. Data-efficient methods in our work help to achieve sustainable storage (also avoid data waste) and energy efficiency. Furthermore, we assess if international and national travel can be avoided by using remote visits. Computational time will be used wisely to avoid too large energy consumptions.

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A120. Willamo, T., Hackman, T., Lehtinen, J. J., **Korpi-Lagg, M. J.**, Kochukhov, O., 2022: "V889: Abrupt changes in the magnetic field or differential rotation?", Open Journal of Astrophysics, https://doi.org/10.21105/astro.2203.13398.

A121. Gent, F. A., Mac Low M.-M., **Korpi-Lagg, M. J.**, and Singh, N. K., 2023: "The small-scale dynamo in a multiphase supernova-driven medium", ApJ, 943, 176. DOI: 10.3847/1538-4357/acac20, arXiv:2210.04460

A122. Warnecke, J., **Korpi-Lagg, M. J.**, Gent, F. A. & Rheinhardt, M., 2023: "Numerical evidence for a small-scale dynamo approaching solar magnetic Prandtl numbers", Nature Astronomy, 7, 662. https://doi.org/10.1038/s41550-023-01975-1

A123. Rempel, M., Bellot Rubio L., Bhatia, T. & **Korpi-Lagg M. J.**, 2023: 'Small-scale dynamos: from idealized models to solar and stellar applications, Space Science Reviews, DOI: 10.1007/s11214-023-00981-z, arXiv: 2305.02787

A124. Yokelson, D., Ramesh, S., Malony, A. D., Norris, B., Lappi, O., Väisälä, M. S., Puro, T., **Korpi-Lagg, M.**, Heljanko, K., 2024: "Observability, Monitoring, and In Situ Analytics in Exascale Applications", CUG2023 conference, in press for the proceedings, and accepted as a journal publication to Concurrency and Computation Practice and Experience.

A125. Snellman, J., Barreiro, N., Barrio, R., Ventura, C., Govezensky, T., Kaski, K., and **Korpi-Lagg, M. J.**, 2024: 'Socio-economic pandemic modelling: Case of Spain', Scientific Reports, 14, 817. https://doi.org/10.1038/s41598-023-44637-y

A126. Gent, F. A., Mac Low, M.-M. & **Korpi-Lagg, M. J.**, 2024: 'Transition from small-scale to large-scale dynamo in a supernova-driven, multiphase medium', ApJ, 961, 7, DOI 10.3847/1538-4357/ad0da0

A127. Hackman, T., Kochukhov, O, Viviani, M., Warnecke, J., **Korpi-Lagg, M. J.**, and Lehtinen, J. J., 2024: 'From convective stellar dynamo simulations to Zeeman-Doppler images', A&A, 12, A156, https://doi.org/10.1051/0004-6361/202347144, arXiv: 2306.07838

A128. **Korpi-Lagg, M. J.**, Mac Low, M.-M., Gent, F. A., 2024: 'Computational approaches to modeling dynamos in galaxies', Living reviews on Computational Astrophysics, 10, 3. https://doi.org/10.1007/s41115-024-00021-9

A129. Snellman, J. E., Barrio, R. A., Kaski, K. K. & **Korpi-Lagg, M. J.**, 2024: "Exploration of the effects of epidemics on the regional socio-economics: a modelling approach", accepted for publication in the Journal of Computational Social Sciences.
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A130. Kivistö, S. K., Gent, F. A., Tornikoski, M., Tammi, J., **Korpi-Lagg, M. J.**, 2024: 'Almost fifty years of Metsähovi solar observations on 37 GHz with recovered digitised historical maps.', under review in A&A, arXiv: 2402.12849

A131. Pekkilä, J., Robertsén, F., Lappi, O., **Korpi-Lagg, M. J.**, 2024: "Stencil Codes on AMD and Nvidia Graphics Processors: Performance and Tuning Strategies", accepted for publication in Concurrency and Computation Practice and Experience. arXiv:2406.08923

A132. Warnecke, J., Viviani, M., Prabhu, A., **Korpi-Lagg, M.**, Rheinhardt, M., 2024: "The role of small-scale dynamos for angular momentum and large-scale dynamo excitation in global convection simulations at high magnetic Reynolds number", accepted for publication in A&A. arXiv:2406.08967

A133. Nguyen, M.-T., Nguyen, A.-D., Rantaharju, J., Puro, T., Rheinhardt, M. **Korpi-Lagg, M. J.**, Truong, H.-L., 'Supporting Opportunistic Data Operations for Data-Intensive Computational Applications', accepted for IEEE Big Data 2024, the 7th Workshop on Benchmarking, Performance Tunin and Optimization for Big Data Applications (BPOD). Dec 15-18, 2024, Washington DC, USA.

A134. Toni, G., Gozaliasl, G. et al. [including Korpi-Lagg, M. J.] 'The COSMOS-Web deep galaxy group catalog up to z=3.7', A&A, in press, arXiv: 2501.09060.

A135. Touko Puro, Matthias Rheinhardt, **Maarit Korpi-Lagg**, Hong Linh Truong, Minh Tri Nguyen, Anh Dung Nguyen 'In-situ and Opportunistic Data Analysis for Exascale Astrophysics Simulations', submitted to Computer physics communications.

B Non-refereed publications in conference proceedings:

B1. **Korpi M. J.**, Brandenburg A., Shukurov A. and Tuominen I. 1998: A local three-dimensional model of supernova regulated ISM; Proceedings of the International Workshop "New Perspectives on the Interstellar Medium", Naramata, British Columbia, Canada, eds. Taylor A. R. and Landecker T. L., PASP Conference Series, 168, 445-448

B2. Tuominen I., Berdyugina S. V., **Korpi M. J.** and Rönty T. 1999: Persistent nonaxisymmteric magnetic structures in late-type stars: observations and models; Proceedings of the International Workshop "Stellar Dynamos: Nonlinearity and chaotic flows", Medina del Campo, Spain, eds. Nunez M. and Ferriz-Mas A., PASP Conference Series, 178, 195-202

B3. **Korpi M. J.**, and Tuominen I. 2003: Stellar dynamos - perspectives and challenges, Magnetism and activity of the Sun and stars: An international conference in honor of the work of Jean-Louis Leroy, eds. J. Arnaud, N Meunier, EDP Publ. Series Vol 9, 9-19 **(an invited review)**

B4. **Korpi, M. J.**, and Mac Low, M.-M. 2003: A numerical investigation on the role of the magnetorotational instability in galactic disks, AN, 324, 67

B5. **Korpi M. J.** 2004: MHD modelling of the turbulent ISM, in the proceedings of the JENAM 2002 ISM Workshop 'From observations to Self-Consistent Modelling of the ISM', eds. M. de Avillez and D. Breitschwerdt, Astrophysics and Space Science, 289, 449-457, Kluwer **(an invited review)**

B6. Mac Low, M.-M., de Avillez, M., **Korpi, M. J.** 2004: The Turbulent Interstellar Medium: Insights and Questions from Numerical Models, in 'How Does the Galaxy Work?', eds. E. J. Alfaro, E. Perez, J. Franco, Astrophysics and Space Science Library, Kluwer, Dordrecht, 315, 339

B7. Tuominen, I., Berdyugina, S. V., **Korpi, M. J.** 2004: Magnetic cycles and rotation in active late-type stars, in Stellar Rotation, Proceedings of the IAU Symposium No. 215, eds. A. Maeder and P. Eenens, 289

B8. Käpylä, P. J., **Korpi, M. J.**, Stix, M., Tuominen I., 2006: Effects of rotation and input energy flux on convective overshooting, in the proceedings of the IAU Symposium 239, "Convection in Astrophysics", eds. F. Kupka, I. Roxburgh and K. Chan, 437-442

B9. Fletcher, A., **Korpi, M. J.**, Shukurov, A. 2009: Dynamically dominant magnetic fields in the diffuse interstellar medium, Cosmic Magnetic Fields: From Planets, to Stars and Galaxies, Proceedings of the International Astronomical Union, IAU Symposium, Volume 259, p. 87-88

B10. Tuominen, I., **Korpi, M. J.**, Käpylä, P. J., Lindborg, M., Ilyin, I. 2009: Stellar nonlinear dynamos: observations and modelling, Cosmic Magnetic Fields: From Planets, to Stars and Galaxies, Proceedings of the International Astronomical Union, IAU Symposium, Volume 259, p. 417-418

B11. Lindborg, M., **Korpi, M. J.**, Tuominen, I., Hackman, T., Ilyin, I., Piskunov, N. 2009: Surface temperature maps for II Peg during 1999-2002, Solar and Stellar Variability: Impact on Earth and Planets, Proceedings of the International Astronomical Union, IAU Symposium, Volume 264, p. 213-218

B12. Brandenburg, A., Käpylä, P. J., **Korpi, M. J.** 2011: From convective to stellar dynamos, in Astrophysical Dynamics: From Stars to Galaxies, proceedings of IAU Symposium 271, eds. N. Brummell and A.S. Brun, Volume 271, p. 279-287

B13. Warnecke, J., Käpylä, P. J., **Mantere, M. J.**, Brandenburg, A. 2012: Coronal ejections from convective spherical shell dynamos, in Comparative Magnetic Minima, proceedings of IAU Symposium 286, p. 154-158

B14. **Mantere, M. J.**, Käpylä, P. J., Pelt, J., 2013: Role of activite longitudes and activity complexes for solar and stellar dynamos, in Solar and astrophysical dynamos and magnetic activity, proceedings of IAU Symposium 294, **(an invited review)**, 175-186

B15. Warnecke, J., Käpylä, P. J., **Mantere, M. J.**, Brandenburg, A. 2013: Solar-like differential rotation and equatorward migration in a convective dynamo with a coronal envelope, in Solar and astrophysical dynamos and magnetic activity, proceedings of IAU Symposium 294, 307-312

B16. Käpylä, P. J., Brandenburg, A., Kleeorin, N., **Mantere, M. J.**, Rogachevskii, I., 2013: Flux concentrations in turbulent convection, in Solar and astrophysical dynamos and magnetic activity, proceedings of IAU Symposium 294, 283-288

B17. Nikbakhsh, S., Tanskanen, E., Hackman, T., **Käpylä, M. J.**, 2018: 'Study of Magnetic Complexity of Solar Active Regions from 1996 to 2017', EGU General Assembly Conference, 20, 16625

C. Scientific books:

C1. Guest editor of the Special issue of Astronomische Nachrichten (Astronomical Notes), Vol. 331, Issue 1, the proceedings of the international conference 'Astrophysical Magnetohydrodynamics', held in the honor of the 50th birthday of Prof. Axel Brandenburg in Kiljava, Finland, during 6.–11.4.2009.

D. Publications intended for professional communities:

D1. Tuominen I., Berdyugina S., Dümmler R. and **Korpi M. J.** 1998: Stellar activity and cosmic magnetic fields; CSC Report on Scientific Computing 1997-1998, ed. J. Haataja, CSC, Finland, 84-89, 197-198

D2. **Korpi M. J.**, Caunt S. E. and Tuominen I. 2000: Modeling MHD phenomena in astrophysical plasmas; CSC News, 12, 6-8

D3. **Korpi M. J.**, Caunt S. E. and Tuominen I. 2001: Turbulence and magnetic fields in the interstellar medium: the role of supernova explosions; CSC Report on Scientific Computing 1999-2000, eds. Sirpa Kotila and Juha Haataja, CSC, Finland, 47-50, 195-196

D4. Käpylä P., **Korpi M. J**. and Tuominen I. 2002: Studying stars and galaxies - a combination of complicated physics and high–performance computing; CSC News, 1/2002, 4-8

D5. Tuominen, I., Hackman, T., Ilyin, I., Kochukhov, O., Käpylä, P. J., Lindborg, M., **Mantere, M. J.**, Piskunov, N. 2011: Azimuthal dynamo wave of an active late-type star detected with SOFIN, NOT Annual Report 2010

D6. **Käpylä, M. J.**, 2018: 'DYNAMO Team Biennial Report 2016-2017 – Part of ReSoLVE Centre of Excellence', Aalto University publication series SCIENCE + TECHNOLOGY, 1/2018.

D7. **Käpylä, M. J.** , Viviani, M., Prabhu, A., Warnecke, J., Duarte, L., Pekkilä, J., Rheinhardt, M.: 'Hunting down the cause of solar magnetism', in "High Performance Computing Science and Engineering, Garching/Munich, 2020", p. 26–28, editors: P. Bastina, D. Kranzmüller, H. Brëuchle, M. Brehm. ISBN: 978-3-9816675-4-7, arXiv:2102.03168

D8. Käpylä, P. J., Brandenburg, A., Gent, Frederick, **Käpylä, M. J.**, Olspert, N., Spada, F., Viviani, M.: 'Cracking the convection conundrum', in "High Performance Computing Science and Engineering, Garching/Munich, 2020", p. 52–56, editors: P. Bastina, D. Kranzmüller, H. Brëuchle, M. Brehm. ISBN: 978-3-9816675-4-7, arXiv:2102.03168

D9. The Astronet Science Vision and Infrastructure Roadmap 2022–2035: Computing; big data, HPC and data infrastructure panel member (**Käpylä, M. J.**), https://www.astronet-eu.org/

E. Publications intended for the general public, linked to the applicant's research

E1. A book chapter about the project 'Supernova-driven interstellar turbulence' of **Korpi, M. J.**, in the book 'Worlds at our Fingertips', written by Markus Nummi, published by the Finnish Culture Foundation, 1999

E2. CSC News -magazine, article about the DYNAMO08 project PId by **Mantere, M. J.**, written by Päivi Brink: Generation of the Sun's magnetic field verified by modeling

E3. Gent, F., **Mantere, M. J.**, Shukurov, A., Fletcher, A., Sarson, G. R., 'Supernova-Regulated Interstellar Medium: Simulations of Disk Galaxies', Science and Supercomputing in Europe - research highlights 2010, 156

E4. **Mantere, M. J.** & Käpylä, P. J.: Saturation of large-scale astrophysical dynamos, DEISA - Advancing Science in Europe, 2011

E5. CSC News -magazine 3/2012, article about the SNDYN project PId by **Mantere, M. J.**, written by Jari Koponen: Magnetic galaxies

E6. Career stories of women working in science and technology, Shaking up Tech event material, Aalto University, 2018; Link to electronic material.

G. Published thesis:

G1. **Korpi M. J.** 1996: Generation of large scale magnetic fields in spiral galaxies: the effect of nonaxisymmetric velocity fields on the dynamo excitation of magnetic fields; Master thesis. Supervisor David Moss (Manchester University), examiner Axel Brandenburg (at that time in the University of Newcastle), grade laudatur (the highest in the Finnish system).

G2. **Korpi M. J.** 1998: Supernova explosions as drivers of interstellar turbulence; Licentiate thesis. Supervisors Axel Brandenburg & Anvar Shukurov (University of Newcastle), examiners Chris Flynn (Tuorla Observatory, Turku) and Dmitry Sokoloff (Moscow University), grade laudatur.

G3. **Korpi M. J.** 1999: Interstellar turbulence and magnetic fields: the role of supernova explosions; Doctoral thesis, University of Oulu. Supervisors Axel Brandenburg & Anvar Shukurov (University of Newcastle), pre-examiners Chris Flynn (Tuorla Observatory, Turku) and Dmitry Sokoloff (Moscow University), opponent in the defense Mordecai-Mark Mac Low (AMNH, New York), kustos in the defense Jukka Jokisaari (University of Oulu), grade laudatur

02.05.2025

The Research Council of Finland

Letter of Collaboration

To whom it may concern,

I am writing to confirm that I have agreed on our research collaboration with the project **"Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)"** lead by Prof. Maarit Korpi-Lagg from Aalto University, submitted to the *"2025 Special funding for international collaboration in high-performance computing"*.

Given the goal to develop AI-assisted simulation tools to model magnetized plasmas, I foresee many challenges in optimizing computing resources, in terms of computational costs and performance, and data resources, in terms of managing and analyzing huge output data volumes. I support the research focus on reducing computational costs, development time, and storage needs. These topics fit very well with my expertise and my team as, over the last many years, I have researched performance observability, cloud and HPC, big data analytics (workflows and streaming analysis) and recently opportunistic data operations optimization in cloud-HPC continuum. Prof. Korpi-Lagg and I were co-PI of the NEOSC subproject in the Inno4scale EuroHPC project, in which my team has developed the opportunistic data analysis platform (ODOP). In this project collaboration, I will support Prof. Korpi-Lagg team to utilize and improve ODOP for MomEnt. Especially, we will not only collaborate on the features of utilizing the idle CPU resources for orchestrating the GPU computations and ML tools for other vital data analysis processes to collect standard data products from the main simulation on the fly, but will also further develop ODOP for versatile and complex HPC workflows in MomEnt.

My team and I will also contribute to techniques and methods for the software prototype and integration of HPC data analysis and resource optimization and workflows. I will have a common supervision of the PhD student and postdoc working on ODOP and HPC workflows. Since Prof. Korpi-Lagg and I are in the same department in Aalto and we have conducted many joint works, I will work closely with Prof. Korpi-Lagg team. I will also provide opportunities for Prof Korpi-Lagg's team members to learn our research activities in our core areas.

Sincerely,

Hong-Linh Truong, Associate Professor

Aalto University
Dept. of Computer Science
Hong-Linh Truong

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P.O. Box 15400
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Konemiehentie 2
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4.5.2025

Letter of Collaboration

This is a letter of collaboration for the Academy Professor project proposal “Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)” of Assoc. Prof. Maarit Korpi-Lagg from Aalto University. I am a Professor of Parallel and Distributed Data Science, specializing in parallel computing methods for big data processing and parallel GPU algorithms. I am very interested in collaboration on GPU-accelerated stencil algorithms, as efficient concurrency control of GPU computing is one of my key research areas. We have already done a joint publication “SOMA: Observability, monitoring, and in situ analytics for Exascale applications” which focused on the analysis on simulation data from massive GPU based HPC simulation using in-memory data streaming between the main simulation and a data analysis application opportunistically using computing CPU resources unused by the GPU simulation. I am very interested in continuing our collaboration also in the context of this new MomEnt proposal, where our main focus will be integration and optimization of the big data analysis workflows to LUMI supercomputer.

Prof. Keijo Heljanko

Deparment of Computer Science

University of Helsinki

Finland

Email: keijo.heljanko@helsinki.fi

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THE UNIVERSITY OF TEXAS AT AUSTIN

Arya Farahi, *Assistant Professor*
Department of Statistics and Data Sciences
E-mail: arya.farahi@austin.utexas.edu
URL: https://afarahi.github.io/

To: Research Council of Finland (RCF)
Project Title: Modelling Turbulent and Magnetised Plasmas at Exascale with AI-assisted Simulation Tools (MomEnt)
Principal Investigator (PI): Prof. Maarit Korpi-Lagg
Call: Special Funding for High-Performance Computing (HPC)

Collaboration Letter

Dear Evaluation Panel,

I am pleased to confirm my collaboration on the *Modelling Turbulent and Magnetised Plasmas at Exascale with AI-assisted Simulation Tools (MomEnt)* project, led by Prof. Maarit Korpi-Lagg. This proposal, submitted to the Research Council of Finland's Special Call for High-Performance Computing, represents a bold and timely effort to push the boundaries of computational astrophysics.

The project addresses one of the most complex challenges in solar and plasma physics: understanding the magnetic dynamo processes of the Sun through the utilization of large-scale, GPU-accelerated simulations. By employing exascale computing, we aim to develop a new generation of AI-powered Plasma simulation that brings new insights into solar magnetism, an understanding essential for predicting Space Weather hazards affecting the Earth. Numerous critical infrastructures, such as satellites, electrical power grids, and communication systems (e.g., navigation via the Global Positioning System; radio communications), remain susceptible to these hazards. Furthermore, in addition to the creation and behavior of plasma observed in natural settings, artificially generated plasmas in laboratories possess a diverse array of technological applications. In particular, Plasma research is crucial in fusion generators where magnetic effects need to be understood to enable the successful production of energy.

Beyond raw computational performance, this project addresses critical challenges in managing petabyte-scale data streams, optimizing AI training workflows, and maintaining a simulation-analysis pipeline that is both high-throughput and interpretable. My group is committed to advancing these areas, aiming to convert computational complexity into new scientific capabilities that align with international priorities in AI, high-performance computing, and solar physics.

I am an Assistant Professor at the University of Texas at Austin and serve as co-PI and AI Lead at the NSF-Simons AI Institute for Cosmic Origins (CosmicAI), established in October 2024. CosmicAI develops advanced AI methodologies to investigate fundamental questions about the origins of the universe. It brings together a consortium of leading institutions, including UT Austin, the University of Utah, the University of Virginia, the National Radio Astronomy Observatory (NRAO), NSF's NOIRLab, Stanford University, and UCLA. The institute's research centers on four core AI themes: trustworthiness, efficiency, interpretability, and robustness. Additionally, it fosters collaboration through open-source tool development, shared data resources, and AI education initiatives aimed at broadening AI accessibility in science communities.

My contributions to this project focus mainly on scalable machine learning, with emphasis on physics-informed neural networks and real-time simulation analysis. In collaboration with Prof. Korpi-Laggs group, we aim to develop AI-driven techniques for efficient interpretation of large-scale solar DNS datasets. A primary goal is the development of robust subgrid-scale models that maintain physical fidelity while reducing computational demands, which is critical for sustainable high-performance computing. Our collaboration will include biweekly virtual meetings, joint development of machine learningaugmented simulation tools, and co-authored publications. Subject to feasibility and funding, we also plan to coordinate research visits and workshops to deepen collaboration. Please note that my participation in this effort does not entail any request for financial support from the RCF.

I support this proposal and look forward to working with Prof. Korpi-Lagg and the Aalto University HPC Lab. Jointly, we have the potential to transform simulations that can potentially revolutionize our understanding of the Sun and push the boundaries of computational astrophysics.

Sincerely,

Prof. Arya Farahi
Assistant Professor
Co-PI, CosmicAI Institute
AI Lead, Explainable Universe

Curriculum Vitae

Maarit J. Korpi-Lagg (aka Korpi, Mantere, Käpylä)
ORCID 0000-0002-9614-2200
April 2025

DEGREES

Nov 2008	**Docent in Astronomy**, Department of Astronomy, University of Helsinki
Sep 1999	**PhD** with honors, Department of Astronomy, University of Oulu, Finland.
Sep 1998	**Licentiate in Philosophy** with honors, Department of Astronomy, University of Oulu, Finland.
Dec 1996	**MSc** with honors, Department of Astronomy, University of Oulu, Finland.

CURRENT EMPLOYMENT

2020 –	**Associate professor (tenured)**, Department of Computer Science, Aalto University, Finland

PREVIOUS WORK EXPERIENCE

2016 – 2024	**Independent Max Planck Research Group Leader**, Max-Planck-Institute for Solar System Research (MPS)/ Germany.
2016 – 2020	**Adjunct professor of Astroinformatics**, team leader of the Centre of Excellence ReSoLVE; School of Science/Department of Computer Science, Aalto University/ Finland
2014 – 2016	**Research Fellow & Academic Coordinator**, ReSoLVE Centre of Excellence DYNAMO team leader, Dept. of Computer Science, Aalto University/Finland.
2013 – 2014	**Research Coordinator**, School of Science, Dept. of Information and Computer Science, Aalto University/Finland.
2012 – 2013	**University Researcher**, Dept. of Physics, University of Helsinki/Finland
2005 – 2012	**Academy Research Fellow**, Dept. of Physics, University of Helsinki/Finland
2005	**Nordita fellow**, Copenhagen/Denmark
2001 – 2004	**Visiting Postdoctoral Fellow** (Academy of Finland mobility grant), Laboratoire d'Astrophysique, Observatoire Midi-Pyrénées, Toulouse/France
2000 – 2001	**Postdoctoral Fellow** (Personal grants from Finnish Culture, Ehrnrooth and Väisälä foundations), University of Oulu/Finland
1997 – 1998	**Visiting Postgraduate Researcher**, University of Newcastle, UK (European Commission Human Capital and Mobility project)
1996 – 1997	**Research Assistant**, Dept. of Astronomy, University of Oulu/Finland

CAREER BREAKS

1999–2009	Altogether 54 months of maternity leaves for the birth of six children.

RESEARCH FUNDING AND GRANTS (MAJOR)

2027 – 2029	Roadmap status in the Finnish infrastructure roadmap and Academy of Finland Funding for the Cosmology Center Finland, PI of Aalto (102,289 euros)
2024 – 2025	EU Inno4scale project 'NEOSC' (200keuros)
2024 – 2026	Academy of Finland 'Finnish Infrastructure' project for Cosmology Center Finland, PI of Aalto University activities. (111,000 euros)
2023 – 2024	ERC Proof-of-Concept Grant "SYCOS" (150keuros).
2021 – 2024	MEC Global US pilot FARIA-program project 'Plasma Physics meets Artificial Intelligence' (140keuros).
2019 – 2024	ERC Consolidator grant 'UniSDyn' (1.964,688 euros)
2018 – 2019	Helsinki Institute for Information and Computer Science (HIIT) seed funding for the EUCLID cosmology mission: big data analysis in virtual supercluster. (21,000 euros)
2016 – 2021	Independent Max Planck Research Group Funding (1,220,000 euros)
2016 – 2020	Academy of Finland 'Finnish Infrastructure Roadmap' project for EUCLID cosmology mission, PI of Aalto University activities. (65,000 euros)
2014 – 2019	Academy of Finland Centre of Excellence project 'ReSoLVE', PI of dynamo team, Aalto University (1,392,062 euros)

2010 – 2013	3-year research breakthrough award for 'Active Suns' project, University of Helsinki, co-PI (560,000 euros)
2005 – 2012	Academy of Finland Research Group funding for Academy Research Fellow (128,000 euros)

RESEARCH OUTPUT:

Open source, user-maintained MHD solver PENCIL CODE (core developer, see A108)
Open source, stencil GPU library Astaroth (project leader, see A115).

RESEACH SUPERVISION AND LEADERSHIP EXPERIENCE

2001 –	Supervised 7 MScs/8 PhDs/8 PDs in astronomy and 13 MScs/2 PhDs (4 ongoing)/2 PDs in computer science.
2016–2024	Leader of the SOLSTAR Max Planck Independent Research Group, MPS, Germany
2020–	Leader of the Aalto HPC Lab and large-scale computing and big data research area, Department of Computer Science, Aalto University, Finland
2014–2019	Leader of the Academy of Finland Center of Excellence 'ReSoLVE' dynamo team, Aalto University, Finland
2017, 2024	Leadership training at MPS and Aalto University.

TEACHING MERITS

2005 –	**Participation in teaching of the astronomy curriculum: Master level and higher**: Stellar structure and evolution, Stellar magnetic activity, Magnetohydrodynamics, Practical Methods in Numerical MHD; **Bachelor level and higher**: Introduction to Astronomy, Introduction to Astronomy: practical exercises, Mathematical Methods in Astronomy. Course materials prepared by the applicant, lectures given at University of Helsinki and Aalto University, Finland.
2010 –	**Guest lecturer in courses belonging to the space physics curriculum**, including Solar Physics (University of Helsinki), and Space Climate (Aalto University).
2010, 2018	**Lecturer in summer/winter schools:** NORDITA winter school on 'Dynamos: above, below, and in the laboratory', CSC-course on 'Complex Flows', CSC summer school on 'Scientific and High-Performance Computing', Universe in a Box, XXXI Canary Islands Winter School, La Laguna, Tenerife.
2020 –	**Design, coordination and supervision of the Data Science Major Bachelor seminar** in the program of Science and Technology, Aalto University.
2020 –	**Participation in teaching and planning of the computer science curriculum: Master level and higher:** Designed and implemented new courses "Programming Parallel Supercomputers" (5cr); "Machine Learning for Universe Sciences" (3cr); "Data-intensive Computing Research project course (5cr).
2020 – 2024	11 credits of pedagogical training for higher education, Aalto University.
2012 – 2013	Studies in **pedagogy, andragogy and didactic of natural sciences, 25 credits**, at the Department of Teacher Education, University of Helsinki
2021–	Academic advisor for Macadamia students in the CCIS Master program, Aalto University
2020–	Academic advisor for Data Science major students in the program of Science and Technology, Aalto University

AWARDS AND HONOURS

2000	Yearly award of the Finnish Academy of Sciences and Letters for high quality PhD thesis
2000	Yearly award of University of Oulu for high quality PhD thesis
1997	Award for the best Master of Science Thesis of the Faculty of Science, University of Oulu

OTHER KEY ACADEMIC MERITS

2004–	30 invited talks at international conferences
2005 –	Participated in organizing 20 international meetings, 7 times as chair of the SOC, 10/3 times as a member of the SOC/LOC.
2022– 2024	Obtained 10 MGPUhours as competitive computing time allocation Europe-wide.
2008–2024	Obtained 220 MCPUhours as competitive computing time allocation Europe-wide.
2023–2025	Participation in the evaluation panel for Vetenskansrådet, Physics of Light and Matter; Astronomy and Subatomic Physics
2022–2023	Member of the Computing panel and participating writing the ASTRONET 2030 roadmap.
2022– 2024	Deputy leader of "Data Science" topic in Helsinki Institute of Information Technology.
2021	Member of the selection committee, Research Fellowship program of the International Research Collaboration Center (IRCC) of NINS, Japan
2020 – 2024	Member of two assistant professor and one lecturer recruitment committees in Aalto University
2020 –	Corresponding fellow at the Nordic Institute of Theoretical Physics, Stockholm, Sweden.
2020 – 2021	Member of the recruitment committee of professors in Space Physics, University of Oulu, Finland
2020	Evaluator for the Polish National Science Centre for their PRELUDIUM program.
2019	Evaluator for NASA ROSES Heliophysics program DRIVE network applications.
2018 – 2020	PhD evaluation committee member of three dissertation (Chi-Ju Wu, Azaymi LitziSiu Tapia, Mariangela Viviani), Georg-August University, Göttingen, Germany
2018	Evaluator for the Swiss National Science Foundation
2017 – 2020	Peer reviewer for RES - Spanish supercomputer network
2016 – 2022	Member of the Steering Group of the Max Planck-Princeton Center for Plasma Physics
2016 – 2024	Member of the advisory group of the directors at MPS, Germany
2013 – 2016	Member of the steering group of Aalto University Large Infrastructures, Aalto University, Finland
2013 – 2016	Member of the steering group of Science-IT, the computational infrastructure of the School of Science, Aalto University, Finland
2013 –	Peer reviewer, EU/PRACE HPC Tier-0 calls
2011 – 2013	Substitute department council member, Dept. of Physics, Univ. of Helsinki/Finland
2010 – 2013	Member of the Physics-Chemistry -network of the Finnish Supercomputing Center (CSC)
2009	Guest editor for Astronomical Notes for the Special Edition 'Astrophysical Dynamos, Axel Brandenburg 50 years'
2001 –	Reviewer for the journals Astronomy & Astrophysics, Astrophysical Journal, Astrophysical Journal Letters, Astronomical Notes, Geophysical and Astrophysical Fluid Dynamics, Entropy, Nature Astronomy, New Journal of Physics, Fluid Dynamics Research, Journal of Plasma Physics, Proceedings of the Royal Society A, Solar Physics, PNAS.

SCIENTIFIC AND SOCIETAL IMPACT

2022–2023	Chair of the departmental Equality, Diversity and Inclusion Committee
2021–	Academic mentor for (three) junior faculty members at the Department of Computer Science, Aalto University.
2005–	Several appearances in public media (newspaper Helsingin Sanomat, Tähdet ja Avaruus, Kaleva, Aamulehti, Prisma Studio in YLE-TV1, YLE Uutiset, PRACE Superdreamers LUMI-G user documentary https://www.lumi-supercomputer.eu/meet-the-super-dreamers/)